# Face/neck leave-on skin care (moisturizers, serums, night creams, leave-on masks): HMTc Standard

Karen Pendergrass · · Heavy Metal Tested & Certified, Institute of Contaminant Standards

CITE AS

Pendergrass, K. (2026). Face/neck leave-on skin care (moisturizers, serums, night creams, leave-on masks): HMTc Standard. Institute of Contaminant Standards.

© 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

## Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for face/neck leave-on skin care (moisturizers, serums, night creams, leave-on masks) in Cosmetics and Personal Care: Leave-on, evaluated on finished leave-on cosmetic, TOTAL CONTENT as placed on market (ug/kg of product as sold), on the technically-unavoidable-impurity basis every instrument in this family uses. Not a migration/leachate basis, not a dietary basis, not an ingredient purity specification.. 5 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

## 1 | How every number on this standard is set

HMTc Method v2.0, government-limit-only initial setting: every numeric cell is the strictest applicable sovereign maximum for this exact leave-on cosmetic form expressed as total content of the finished product as placed on market, or a disclosed read-across where the exact cell is unregulated, or a substantiated not-material determination, or a control. No cell is occurrence-derived, no cell is precautionary or feasibility-adjusted, and no cell is anchored on an ingredient or colour-additive purity specification.

## 2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 1,000 | government maximum level |
| Cadmium (Cd) | Tier 1 | 100 | government maximum level |
| Total arsenic (tAs) | Tier 2 | 500 | government maximum level |
| Inorganic arsenic (iAs) | Tier 1 | ≤ tAs | bounded by tAs (reflex speciation) |
| Mercury, total (tHg) | Tier 1 | 100 | government maximum level |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 10,000 | government maximum level |
| Tin (Sn) | Tier 2 | 3,750,000 | human-grade read-across (European Union) |
| Aluminium (Al) | Tier 2 | 560,000 | human-grade read-across (European Union) |
| Total chromium (Cr) | Tier 2 | 50 | human-grade read-across (AU) |
| Hexavalent chromium (Cr-VI) | Tier 2 | 5 | human-grade read-across (European Union) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 1,000 | government maximum level |
| Cadmium (Cd) | 100 | government maximum level |
| Total arsenic (tAs) | 500 | government maximum level |
| Inorganic arsenic (iAs) | ≤ tAs | iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell. |

| | | |
|---|---|---|
| Mercury, total (tHg) | 100 | government maximum level |
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 10,000 | government maximum level |
| Tin (Sn) | 3,750,000 | European Union — human-food analog |
| Aluminium (Al) | 560,000 | European Union — human-food analog |
| Total chromium (Cr) | 50 | AU — human-food analog |
| Hexavalent chromium (Cr-VI) | 5 | European Union — human-food analog |

## 4 | Cite this standard

Pendergrass, K. (2026). *Face/neck leave-on skin care (moisturizers, serums, night creams, leave-on masks): HMTc standard*. Institute of Contaminant Standards.

© 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

## References

[1] AU — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Sn, Al, Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

### STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

### DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at

### PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/face-neck-leave-on-skin-care.

https://heavymetalcertified.com/standards/face-neck-leave-on-skin-care. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

© All rights reserved